100 Days of Linux - Day 063

Title: Working with Variables in Bash Scripts

Today's Goal
Learn how to create, use, and reuse variables inside Bash scripts.

Why This Matters
Variables make shell scripts flexible and reusable. They are used in automation, DevOps, ETL pipelines, backups, and deployment scripts.

Commands & Concepts of the Day

Create a Variable:
NAME="Vishal"
echo $NAME

Built-in Variables:
USER, HOME, PWD

Command Substitution:
TODAY=$(date)
CURRENT_DIR=$(pwd)

Example Script:
#!/bin/bash
NAME="Linux Student"
echo "Welcome $NAME"
echo "Current Directory: $(pwd)"

Beginner Lesson
A variable stores a value that can be reused throughout your script. In Bash, do not put spaces around the '=' sign. Use the dollar sign ($) to read a variable's value.

Practice Questions

1. Create a script named variables.sh that stores your name in a variable and prints it.
2. Add another variable named COURSE with the value Linux100 and display both variables.
3. Store today's date in a variable using $(date) and print it.
4. Store your current working directory using $(pwd) in a variable and display it.
5. Run the script and then display your current working directory.

Hints
Remember: VARIABLE=value (no spaces). Use $VARIABLE to display its value.

Mini Challenge
Create student_info.sh that stores your name, course, current directory, current date, and current user in variables, then prints a

formatted report.

Common Mistakes
Adding spaces around '=' or forgetting the '$' when displaying a variable.

Did You Know?
Most production shell scripts rely heavily on variables to avoid hard-coding values such as file paths and usernames.

Pro Tip
Use meaningful variable names like PROJECT_NAME or BACKUP_DIR instead of single letters.

Answer Key (Instructor Only)
Q1: NAME="YourName" && echo $NAME
Q2: COURSE="Linux100" && echo $NAME $COURSE
Q3: TODAY=$(date) && echo $TODAY
Q4: CURRENT_DIR=$(pwd) && echo $CURRENT_DIR
Q5: bash variables.sh && pwd

Homework
Create three scripts that use variables for names, directories, and dates. Practice modifying only the variables while keeping the rest of the script unchanged.

Next Day Preview
Tomorrow you'll learn how to accept user input in Bash scripts using the read command.